
Problem 9.16. Prove that $TQBF \notin \text{SPACE}(n^{1/3})$.

Proof. We know that

$$TQBF \in \text{PSPACE},$$

and

$$\text{SPACE}(n) \subseteq \text{PSPACE}.$$

According to Space hierarchy theorem

$$\text{SPACE}(n^{1/3}) \subsetneq \text{SPACE}(n).$$

Therefore,

$$TQBF \notin \text{SPACE}(n^{1/3}).$$

□